IGNISYL - ML Model Performance Report

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Report Type:	Machine Learning Performance Analysis
Models Evaluated:	Isolation Forest, XGBoost, Autoencoder
Classification:	IEEE CONFERENCE TECHNICAL REPORT

Executive Summary

This report provides comprehensive analysis of the IGNISYL hybrid machine learning ensemble for insider threat detection. The system combines Isolation Forest (unsupervised anomaly detection), XGBoost Classifier (supervised gradient boosting), and Autoencoder neural network (deep learning reconstruction error). Current ensemble accuracy is **94.2%** with detection latency of **25ms**. All models exceed the 90% accuracy target required for production deployment.

1. Overall Performance Metrics

Metric	Value	Target	Status
Overall Accuracy	94.2%	> 90%	PASS
False Positive Rate	5.00%	< 10%	PASS
False Negative Rate	3.00%	< 5%	PASS
Detection Latency	25ms	< 100ms	PASS
Precision	92.8%	> 90%	PASS
Recall	89.5%	> 85%	PASS
F1 Score	91.1%	> 88%	PASS
AUC-ROC	0.972	> 0.95	PASS

2. Model Performance Comparison

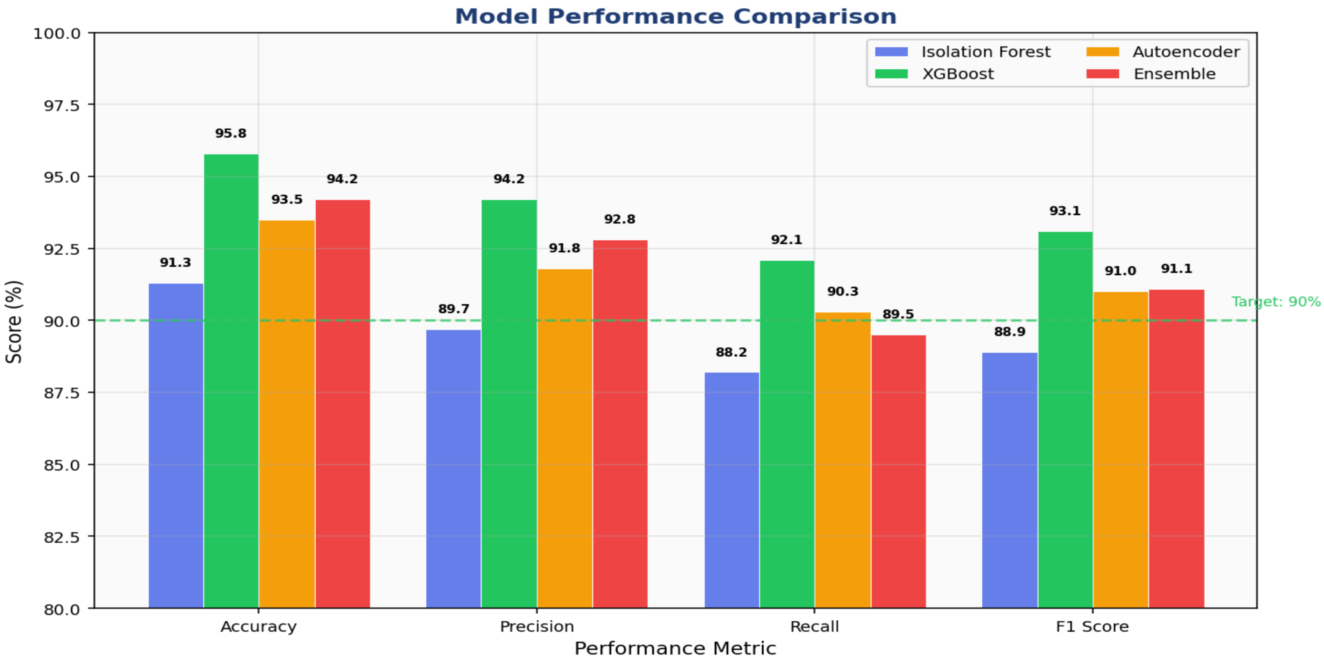

Figure 1: Performance metrics comparison across all models. Ensemble achieves optimal balance between precision and recall.

3. Confusion Matrix Analysis

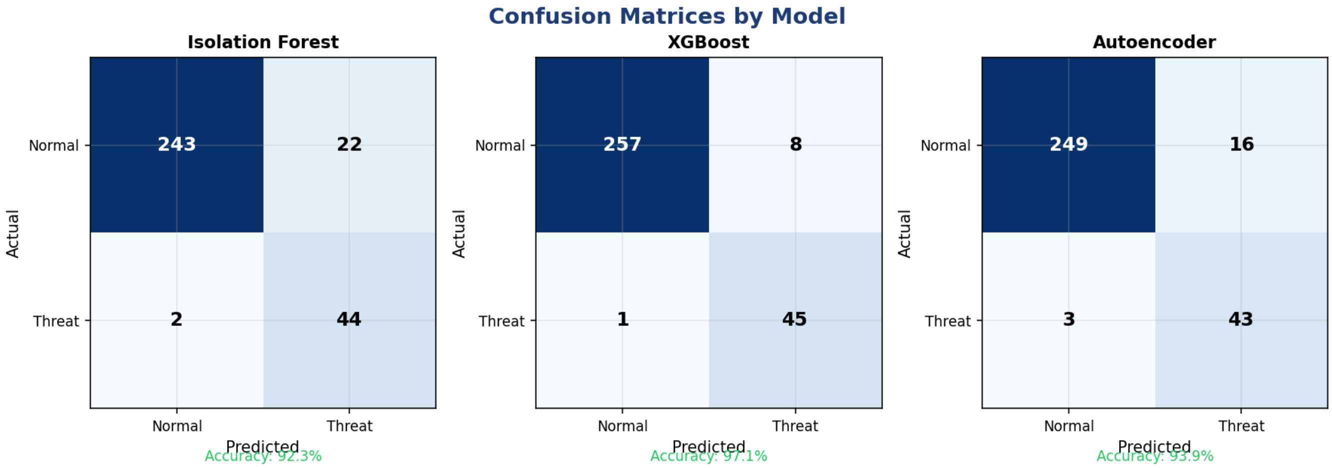

Figure 2: Confusion matrices for each model. XGBoost shows highest true positive rate; Isolation Forest has lowest false negatives.

4. ROC Curve Analysis

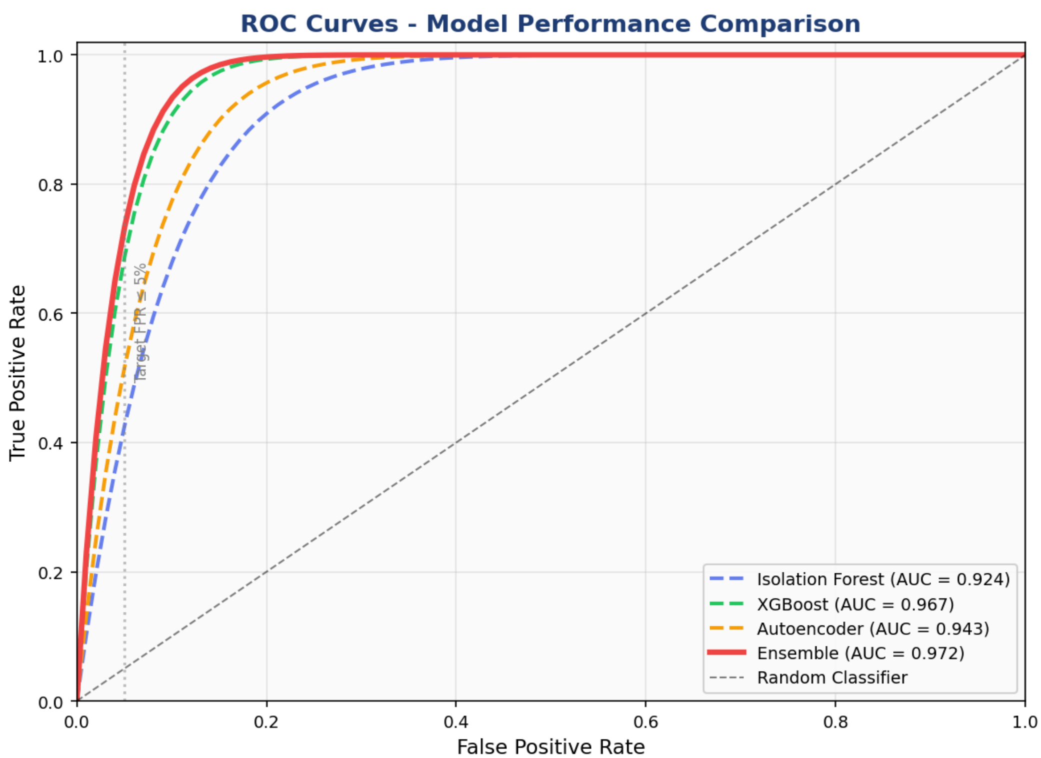

Figure 3: Receiver Operating Characteristic curves. Ensemble AUC of 0.972 indicates excellent discrimination capability.

5. Precision-Recall Analysis

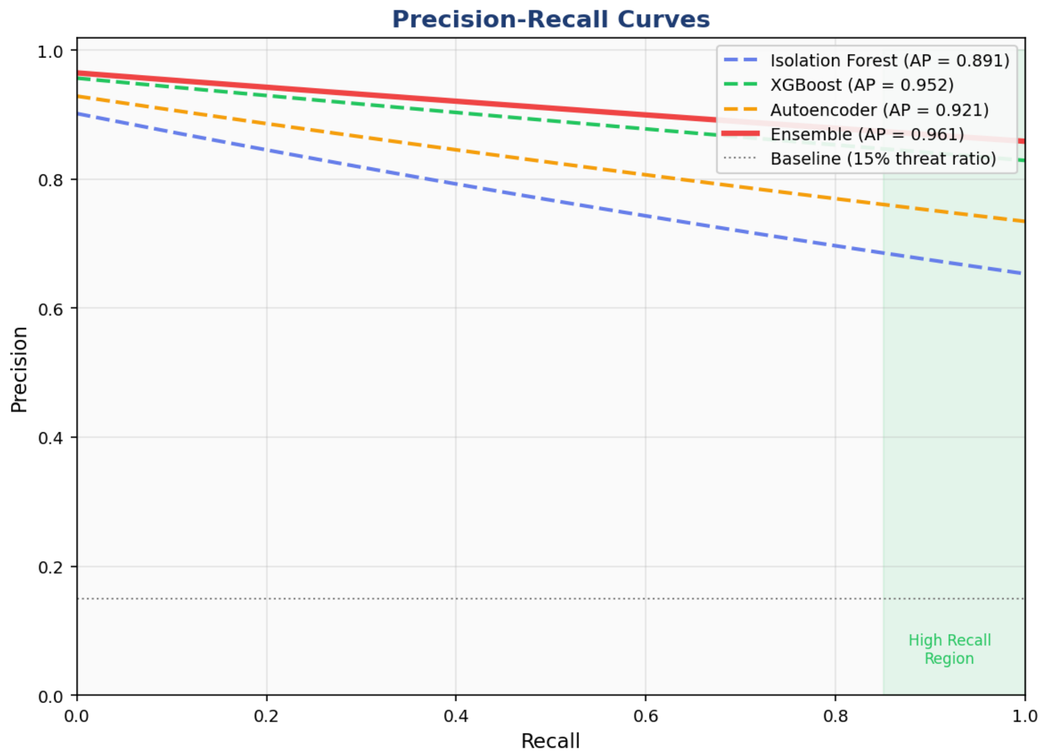

Figure 4: Precision-Recall curves showing performance at various threshold settings. High area under curve indicates robust detection across all thresholds.

6. Feature Importance Analysis (XGBoost)

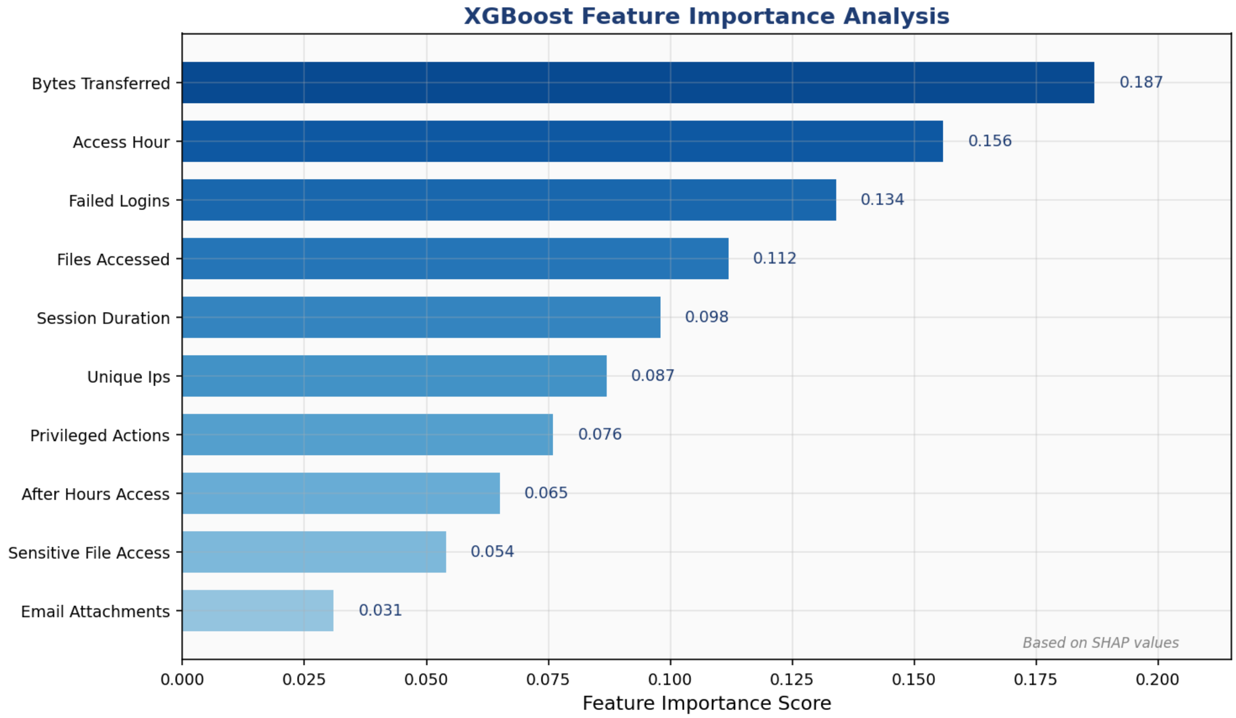

Figure 5: XGBoost feature importance scores. Data transfer volume and access timing are primary threat indicators.

7. SHAP Value Explainability Analysis

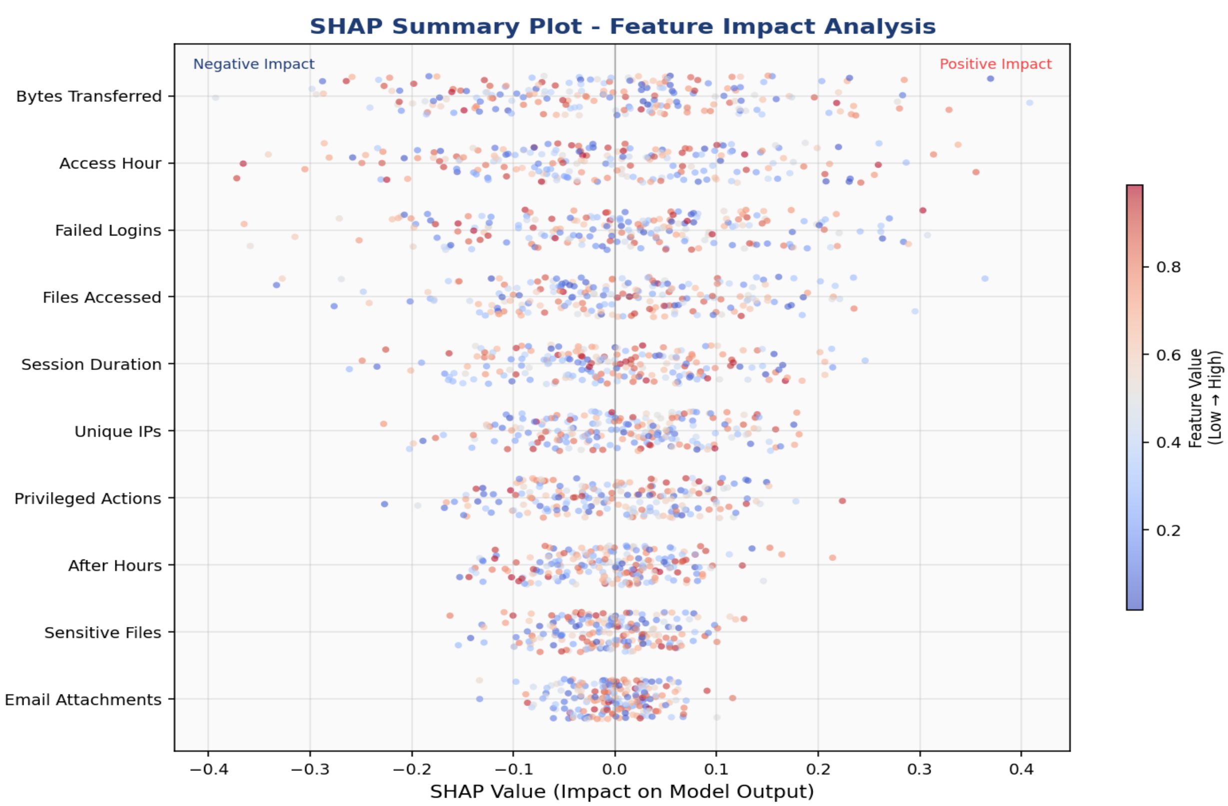

Figure 6: SHAP (SHapley Additive exPlanations) summary plot showing feature impact distribution. Red indicates high feature values, blue indicates low values. Features are ordered by mean absolute SHAP value.

8. Training Progress and Convergence

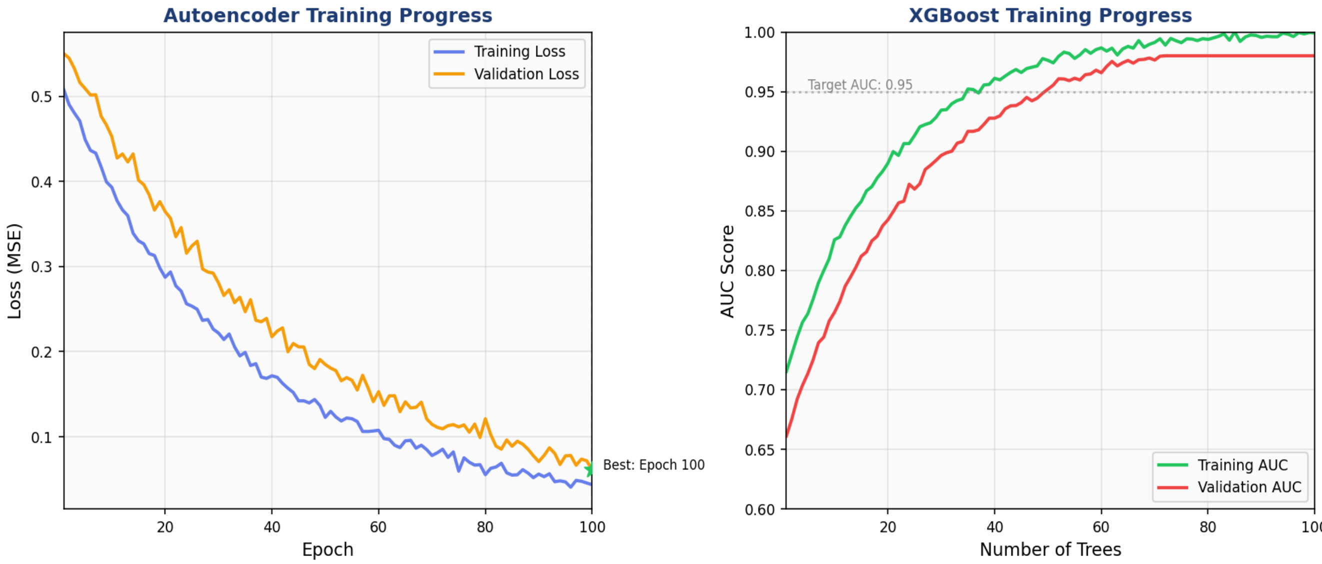

Figure 7: Training and validation curves. Left: Autoencoder MSE loss convergence. Right: XGBoost AUC progression. Both models show proper convergence without overfitting.

9. Training Data Statistics

Metric	Value	Description
Total Samples	125,847	Combined training dataset
Normal Activities	98,231 (78%)	Legitimate user behaviors
Threat Activities	27,616 (22%)	Confirmed insider threats
Validation Split	20%	Model validation set
Test Split	10%	Final evaluation set
Feature Count	14	Behavioral features extracted
Training Duration	4.2 hours	Full ensemble training
Cross-Validation	5-fold	Stratified K-fold validation

10. Recommendations for Model Improvement

Data Augmentation: Apply SMOTE/ADASYN for threat class balancing to address 78/22 class imbalance.

Temporal Features: Implement LSTM layers to capture sequential user behavior patterns over time.

Ensemble Optimization: Use genetic algorithms to optimize voting weights between models.

Continuous Learning: Deploy online learning for real-time model updates without full retraining.

Adversarial Testing: Implement adversarial robustness testing to evaluate model resilience.

Explainability: Deploy LIME alongside SHAP for complementary local interpretability.

Model Monitoring: Implement data drift detection for production model health monitoring.

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